

Apratim Das

Robotics — AI — Embedded Systems — Full-Stack — Creative Tech

B.Tech Electrical and Electronics Engineering, IIT Dharwad (2024–2028)

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Education

Indian Institute of Technology Dharwad

2024–2028

B.Tech in Electrical and Electronics Engineering

Experience

Robotics & AI Industrial Trainee, My Equation

2025

- Designed ESP32-based obstacle avoidance robot using ultrasonic sensing
- Built 6-DOF robotic manipulator using URDF and integrated with ROS2, RViz
- Simulated industrial robotic workflows using ABB RoboStudio and AutoCAD

Virtual Tech Consultant (Simulation), Deloitte Australia

2025

- Built Python-based IIoT telemetry pipeline for real-time industrial monitoring
- Standardized multi-source industrial data and designed monitoring dashboards

Music Director & Video Engineer, 1NOT2 Productions

2026–Present

- Led music production, audio engineering, and creative direction
- Managed end-to-end video production workflows and editing pipelines

Freelance Video Editor & Content Strategist

2025–Present

- Produced high-engagement digital content (300K+ reach)
- Designed storytelling-driven campaigns combining analytics and creativity
- Designed official merchandise for IIT Dharwad contingent (Inter IIT Cultural Meet 8.0, IIT Kanpur)

Campus Ambassador – Techniche IIT Guwahati

2025

- Received Letter of Recommendation for leadership and professional responsibility

Campus Ambassador – Google Gemini & Paytm

2026–Present

- Driving product outreach, awareness, and campus engagement initiatives

Projects

BRAVO – AI Companion Robot (ESP32 + OpenAI + Web Control)

- Built ESP32-based robot with real-time video streaming and web-based control
- Integrated OpenAI API for conversational interaction
- Implemented human-following system using Python-based vision tracking
- Developed WebSocket-based remote control interface

Aquifer Sentry – Smart Water Monitoring System (IIT Madras Finalist)

- Designed IoT system to detect illegal groundwater extraction
- Applied continuity equation for anomaly detection in flow systems
- Built ESP32-based real-time sensing system for water monitoring

Autonomous Line Following Robot (RoboRace 2025)

- Built high-speed robot using Arduino Nano and IR sensor array
- Implemented real-time control logic for dynamic path correction

Industrial Palletizing Simulation (ABB IRB 4600)

- Designed robotic palletizing system with dual infeed conveyors
- Simulated industrial pick-and-place operations with motion planning

Obstacle Avoidance Robot

- Built embedded robotic system using ESP32 and ultrasonic sensors

Leadership & Positions

Team Lead – S2C2 Smart City Challenge (IIT Madras)

Team Lead – Empyrean Nexus (Build-a-Bot, PARSEC IIT Dharwad)

Team Lead – Mauryan Minds (Nasiko AI Hackathon)

- Led robotics and AI teams to Top 15 national rankings (300+ teams)

Event Lead – Beyond the Roll (PARSEC 2026)

- Organized large-scale inter-college B-Roll filmmaking competition
- Managed end-to-end execution, judging, and multi-institute participation

Marketing & Video Editing Wing Lead

- PARSEC 2026, HarshTaal 2025, E-Summit 2024 & 2025
- Led branding, content production, and social media campaigns

Council Member – Photography & Films Club, IIT Dharwad

Active Member – Coding Club, AI Club, Electronics Club, Music Club (IIT Dharwad)

Achievements

- Top 5 – Build-a-Bot Competition (PARSEC IIT Dharwad)
- Top 15 – S2C2 Challenge (IIT Madras, 300+ teams)
- Top 10 – Nasiko AI Hackathon
- 1st Prize – B-Roll Ad Making Competition
- 1st & 3rd – Fashion Photography Competitions

Skills

Programming: Python, C, JavaScript, DSA

Robotics/Systems: ESP32, Arduino, Embedded Systems, ROS2 (basics)

AI/ML: OpenAI API, Computer Vision

Tools: ABB RoboStudio, AutoCAD, Wokwi, RViz, MATLAB, IDE

Web: HTML, CSS, WebSockets

Creative Tools: DaVinci Resolve, Adobe Lightroom, Canva

Other: Video Editing, Music Production, Digital Marketing

Certifications

- AWS Solutions Architecture Job Simulation (Forage)
- Deloitte Technology Job Simulation
- Robotics & AI Industrial Training (My Equation)